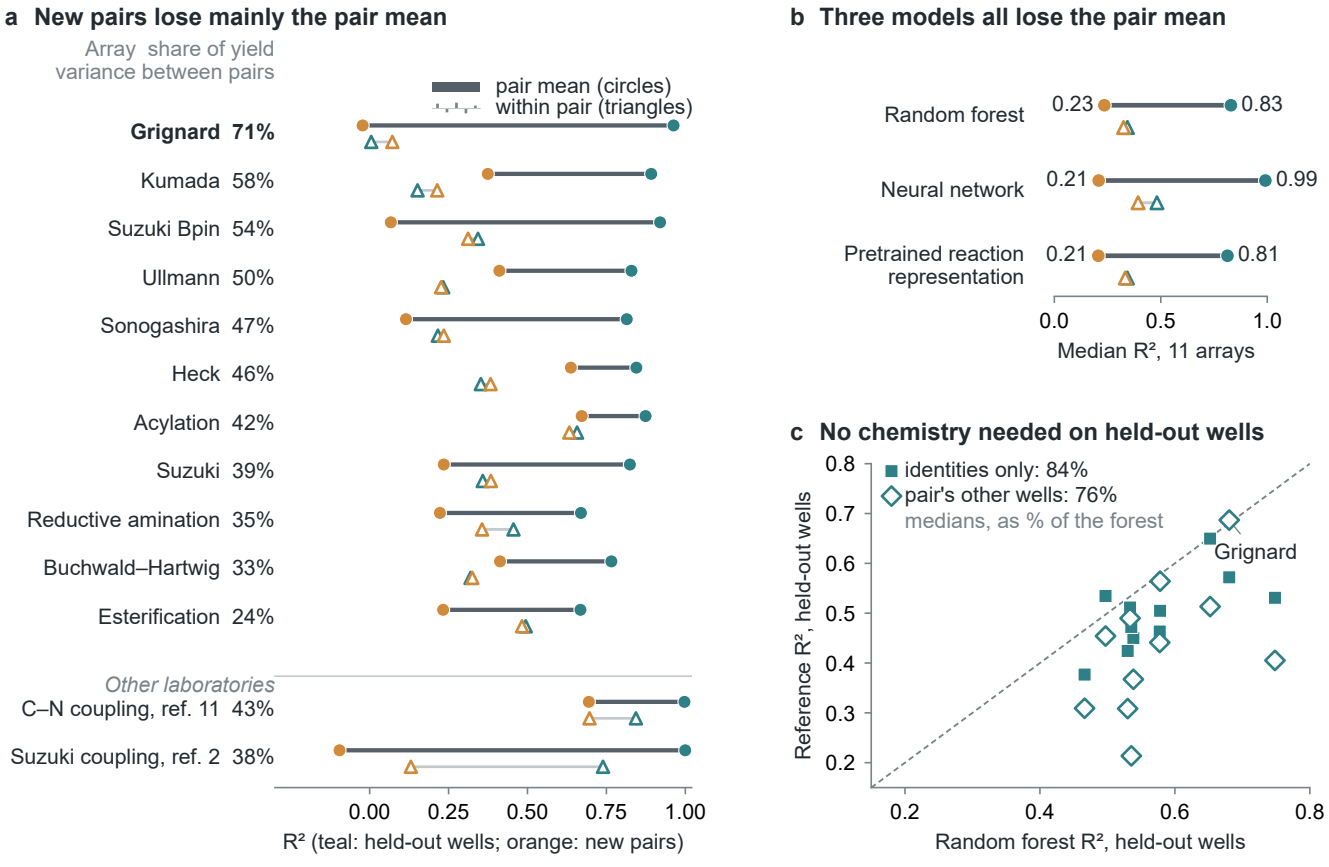

a New pairs lose mainly the pair mean
Array share of yield variance between pairs
pair mean (circles)
within pair (triangles)
Grignard 71%
Kumada 58%
Suzuki Bpin 54%
Ullmann 50%
Sonogashira 47%
Heck 46%
Acylation 42%
Suzuki 39%
Reductive amination 35%
Buchwald–Hartwig 33%
Esterification 24%
Other laboratories
C–N coupling, ref. 11 43%
Suzuki coupling, ref. 2 38%
0.00 0.25 0.50 0.75 1.00
R² (teal: held-out wells; orange: new pairs)
b Three models all lose the pair mean
Random forest 0.23 0.83
Neural network 0.21 0.99
Pretrained reaction representation 0.21 0.81
0.0 0.5 1.0
Median R², 11 arrays
c No chemistry needed on held-out wells
identities only: 84%
pair's other wells: 76%
medians, as % of the forest
Grignard
Reference R², held-out wells
0.2 0.3 0.4 0.5 0.6 0.7 0.8
0.2 0.4 0.6 0.8
Random forest R², held-out wells